

Digital Advertising Performance Across Platforms x Campaign Type (Google, Meta, TikTok Ads)

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Introduction

Research Questions

- Do Google Ads, Meta Ads, and TikTok Ads produce structurally different campaign performance patterns after accounting for campaign context?
- How can those platform-level differences inform budget allocation when advertising returns diminish at higher levels of spend?

Why This Topic Matters

- “Enterprises seek to harness the power of SM as a strategic tool to improve firm value, profits, and competitive advantage” (Ahmad et al.).
- “SM is a useful tool that could help businesses get closer to the customers and become more competitive” (Ahmad et al.).
- “Marketing experts need to think strategically, integrate social media campaigns into their marketing plans and use the right tools in the right way” (Bekoglu and Onaylı).

Data

Global Advertising Campaign Dataset

- **Source:** Kaggle global ad performance dataset
- **Rows:** 1,800 campaign observations
- **Platforms:** Google Ads, Meta Ads, TikTok Ads
- **No missing values** in the analysis file
- **Key variables:** impressions, clicks, CTR, CPC, conversions, CPA, ad spend, revenue, and ROAS

Variable Groups Used in the Analysis

- **Exposure:** impressions, clicks
- **Efficiency:** CTR, CPC, CPA, ROAS
- **Financial outcomes:** ad spend, revenue, conversions
- **Context variables:** campaign type, industry, country

Methods

Methods: Analytical Roadmap

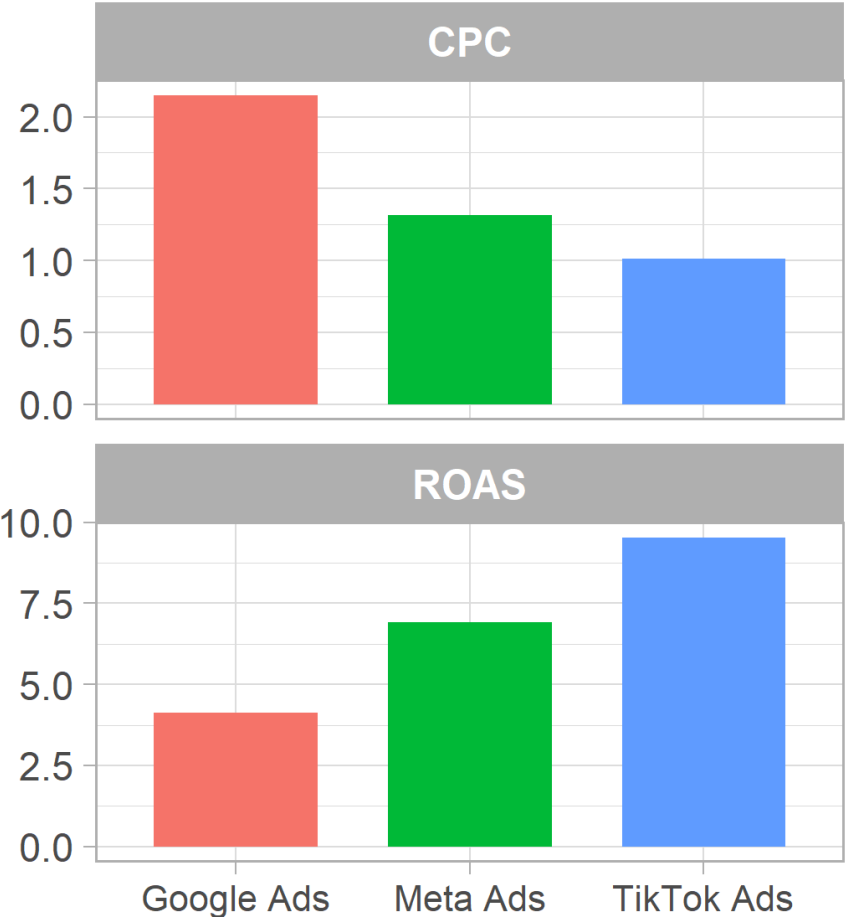
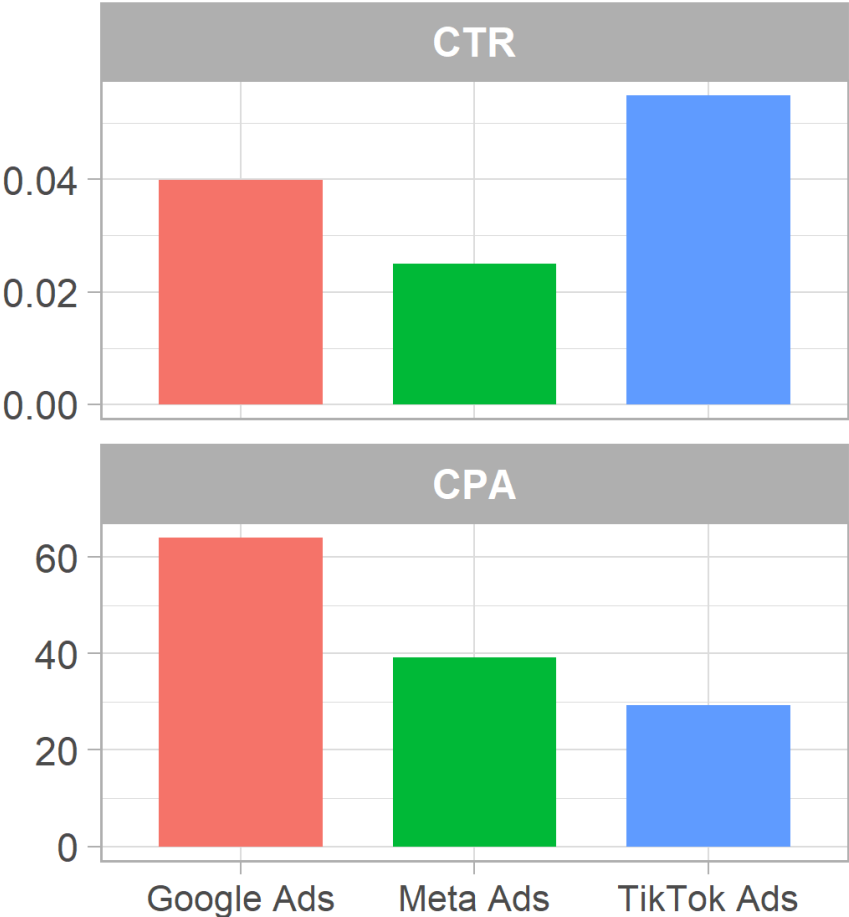
- Descriptive summaries
- Multiple Factor Analysis (MFA)
- ANOVA + emmeans
- K-means clustering

Methods: Why These Methods

- MFA → handles correlated metrics
- ANOVA → tests platform differences across each MFA dimension- Regression → detects diminishing returns
- Optimization → converts model into decisions
- Random forest → captures nonlinear patterns

Results

Average Platform Metrics for CTR, CPC, CPA, and ROAS



Distribution of ROAS by Campaign Type

50



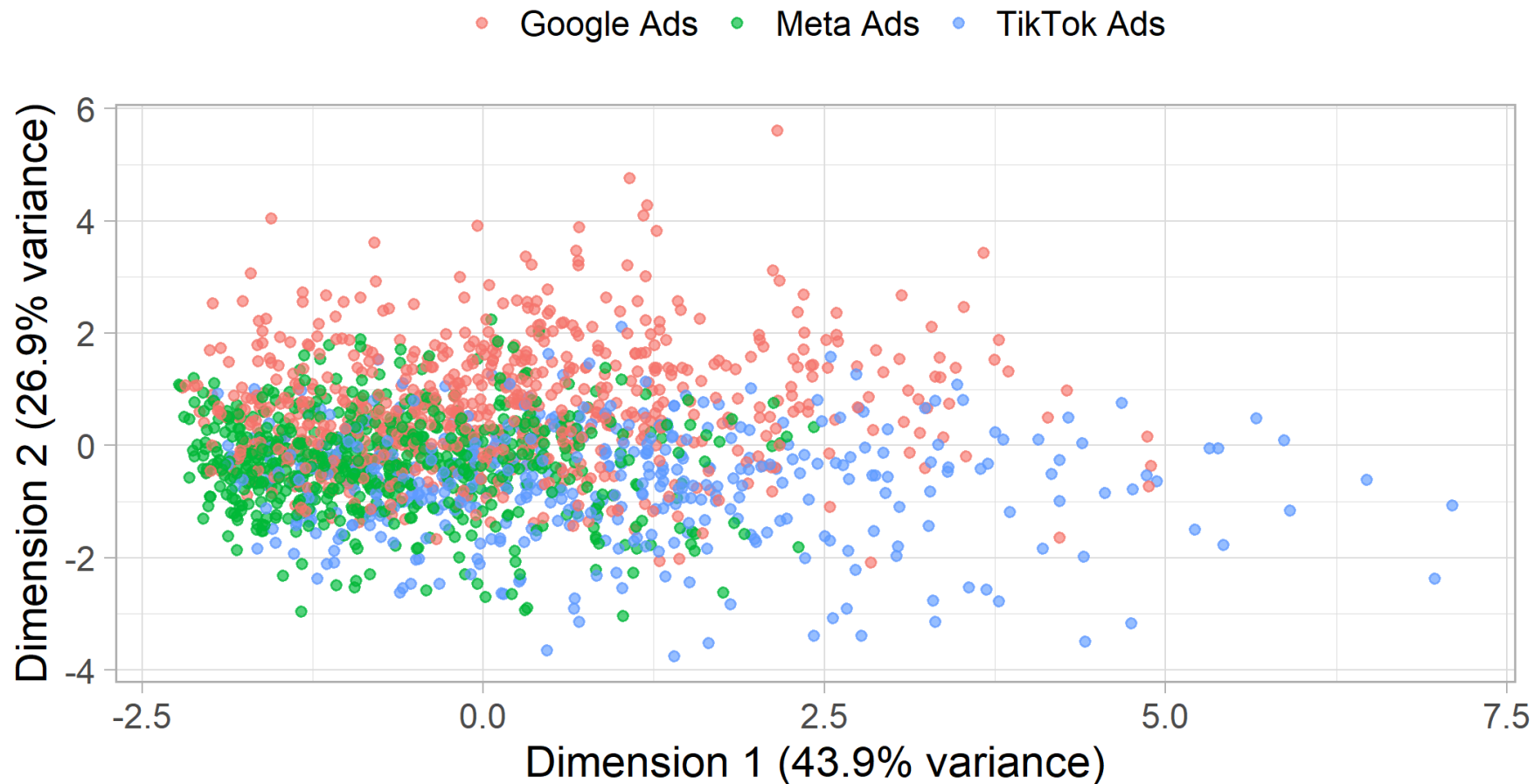
Multiple Factor Analysis (MFA)

- Combines many correlated performance metrics
- Accounts for groups of variables (exposure, efficiency, financial)
- Reduces data into a few dimensions

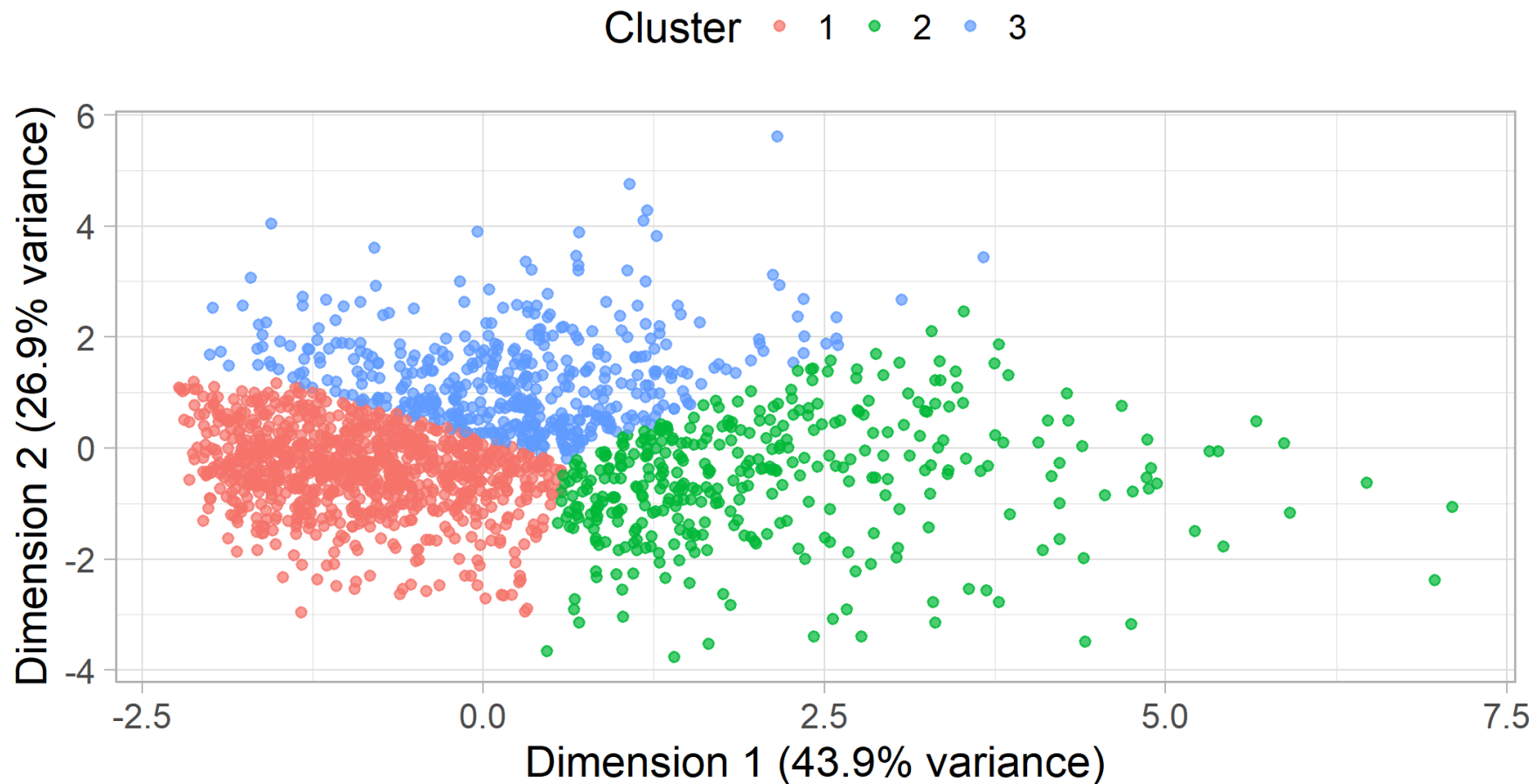
Output:

- Each campaign is represented as a point
- Dimensions summarize overall performance patterns

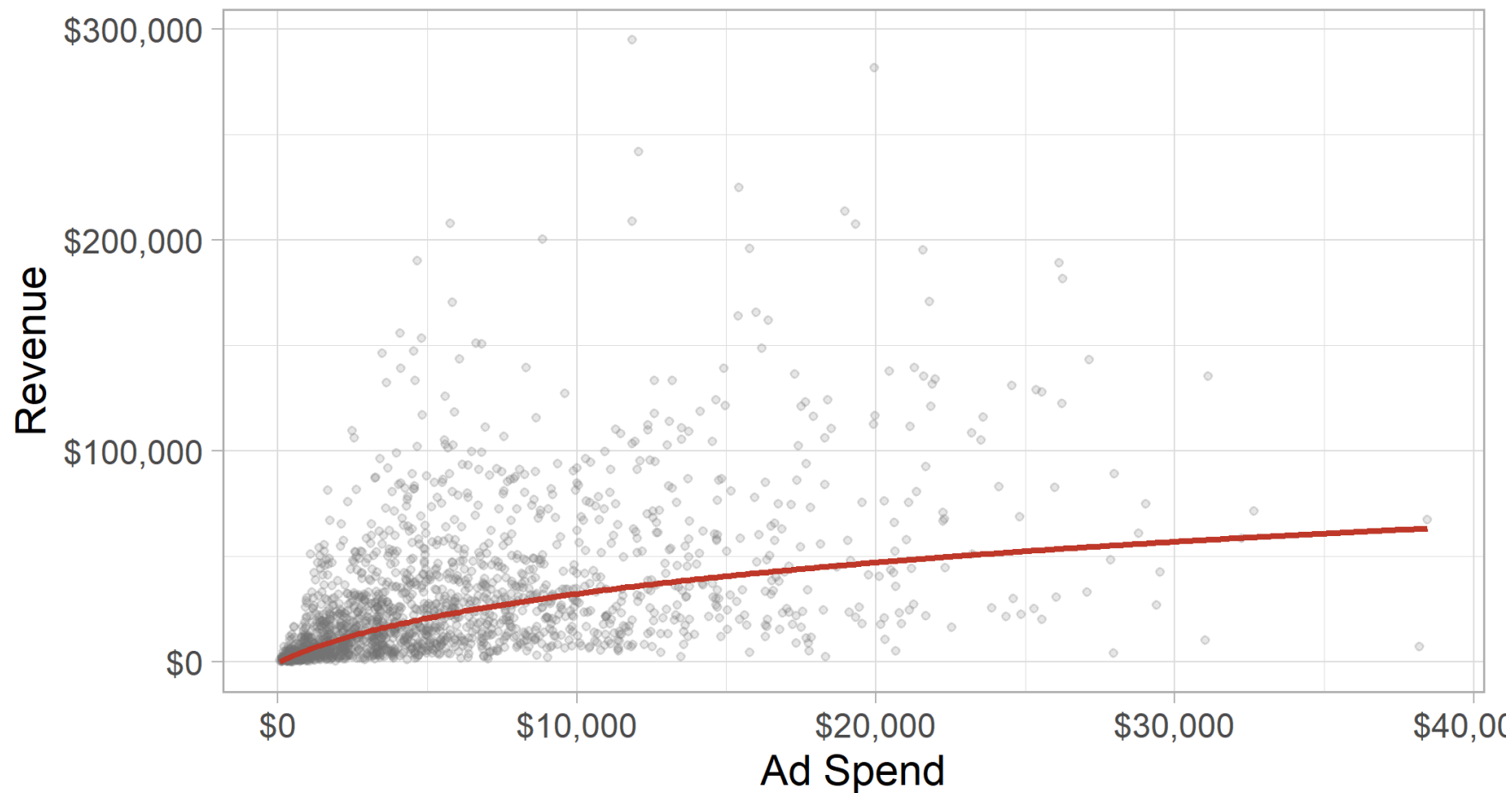
Campaigns in the First Two MFA Dimensions, Colored by Platform



K-Means Clustering of Campaigns in MFA Space



Observed Revenue and the Fitted Spend–Revenue Curve



How Budget Optimization Works

- We model how revenue changes with ad spend
- Revenue increases, but with **diminishing returns**
- Instead of splitting budget evenly, we allocate step-by-step

Key idea:

At each step, assign budget to the platform with the **highest marginal gain in revenue**

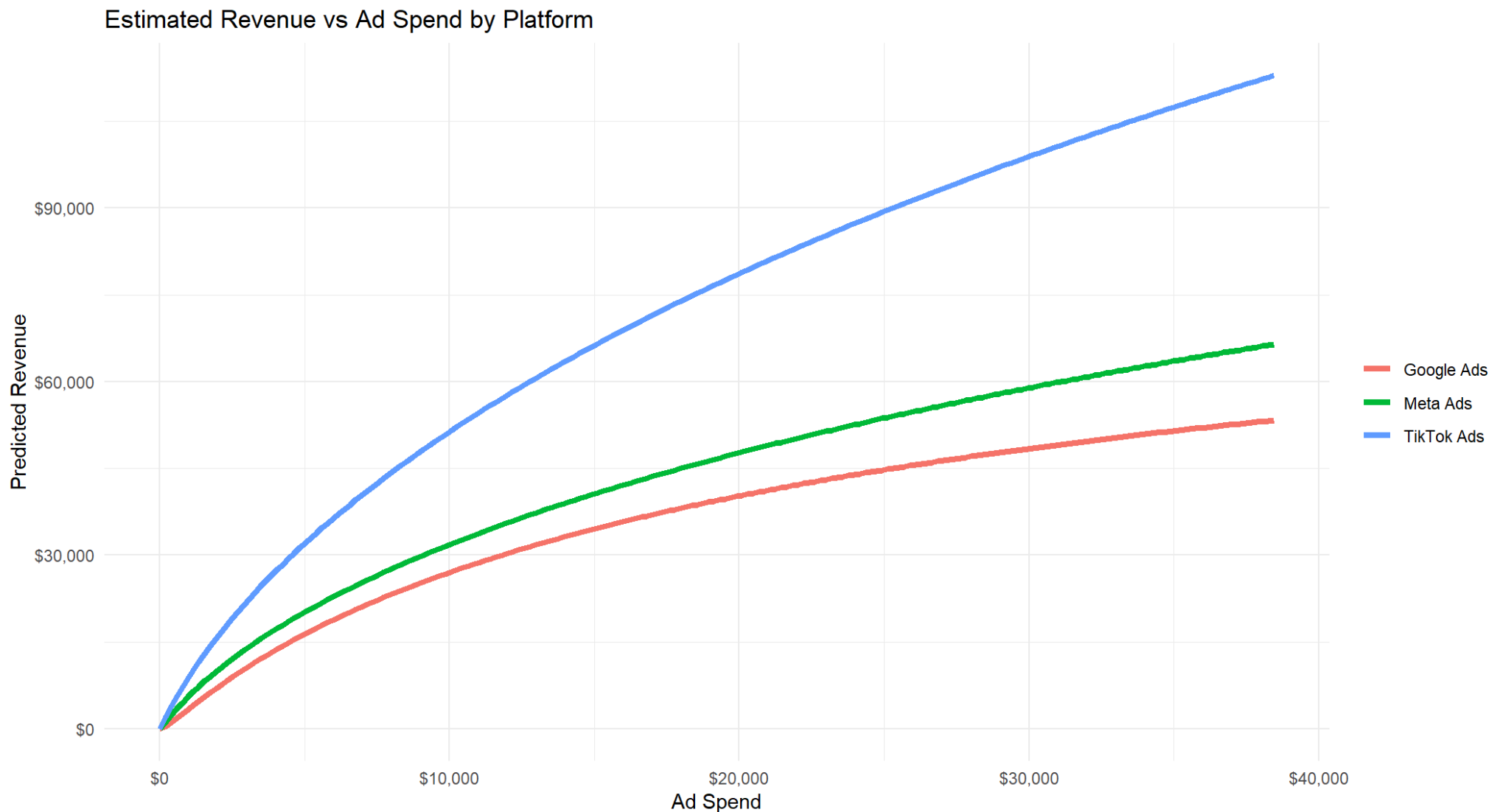
Equation

Modeling Revenue as a Function of Ad Spend

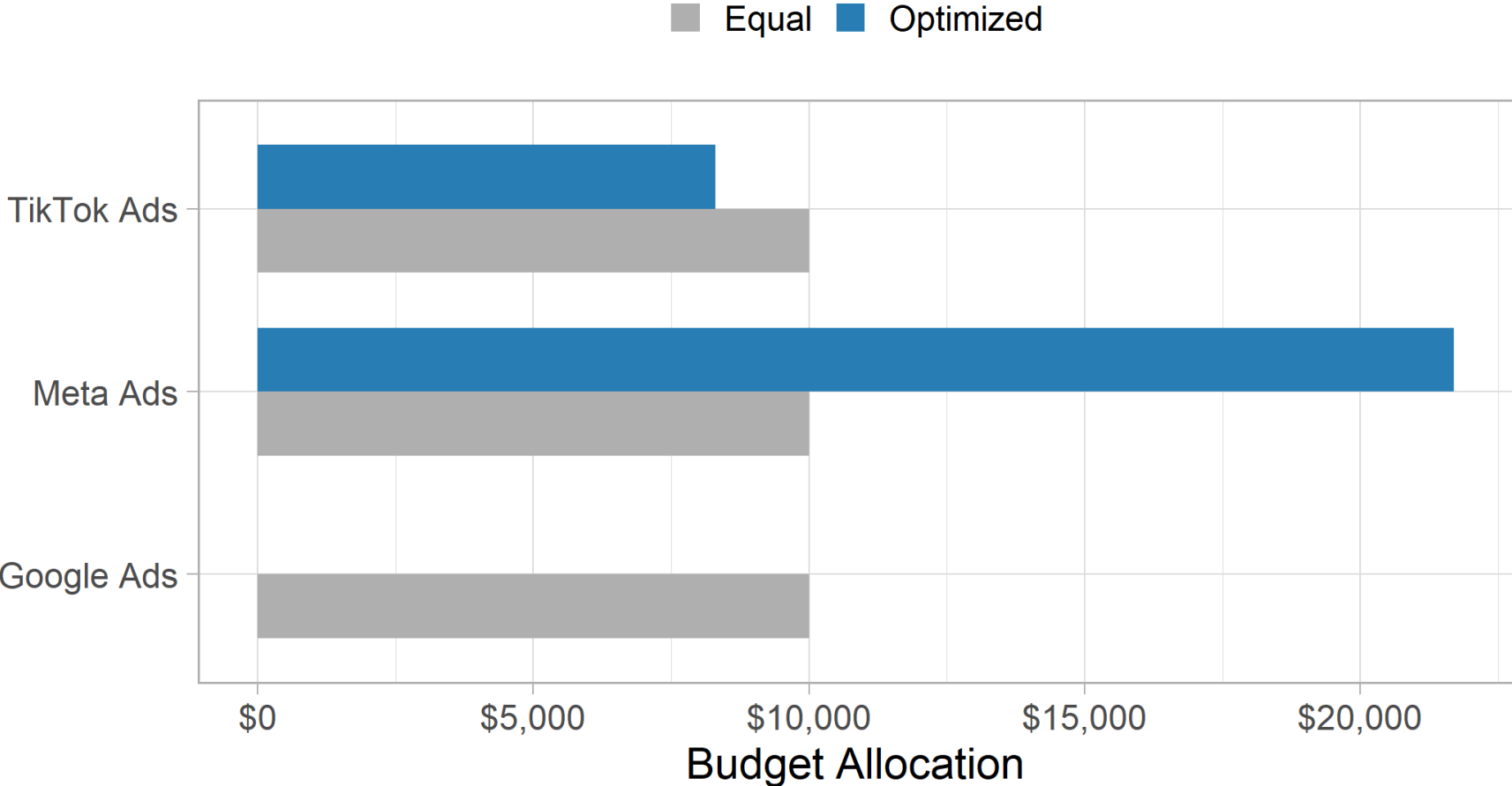
$$\log(\text{Revenue} + 1) = \beta_0 + \beta_1 \log(\text{Ad Spend} + 1) + \beta_2 [\log(\text{Ad Spend} + 1)]^2$$

• $\log(\text{Demand}_{i,t} + 1)$: stabilize variation and handle

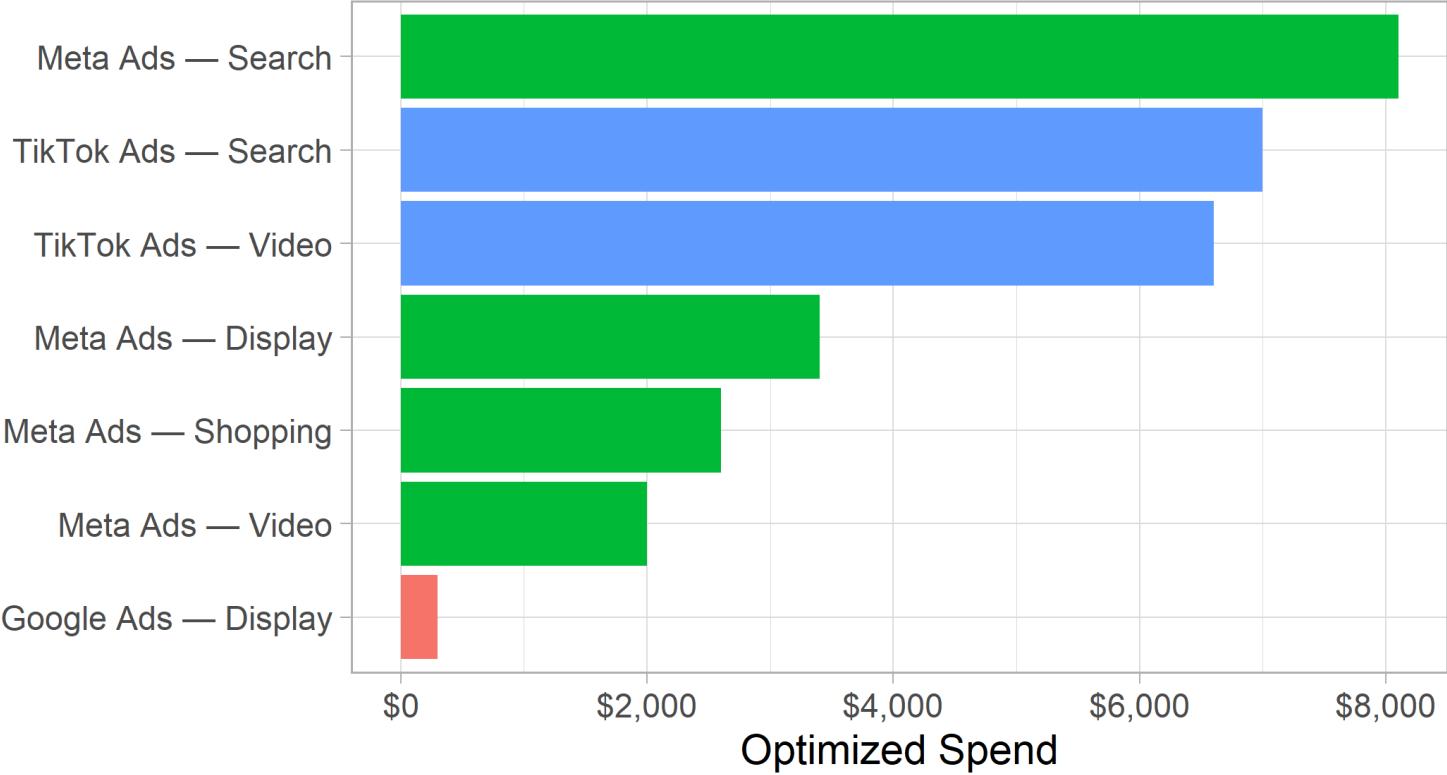
Diminishing Returns by Platform



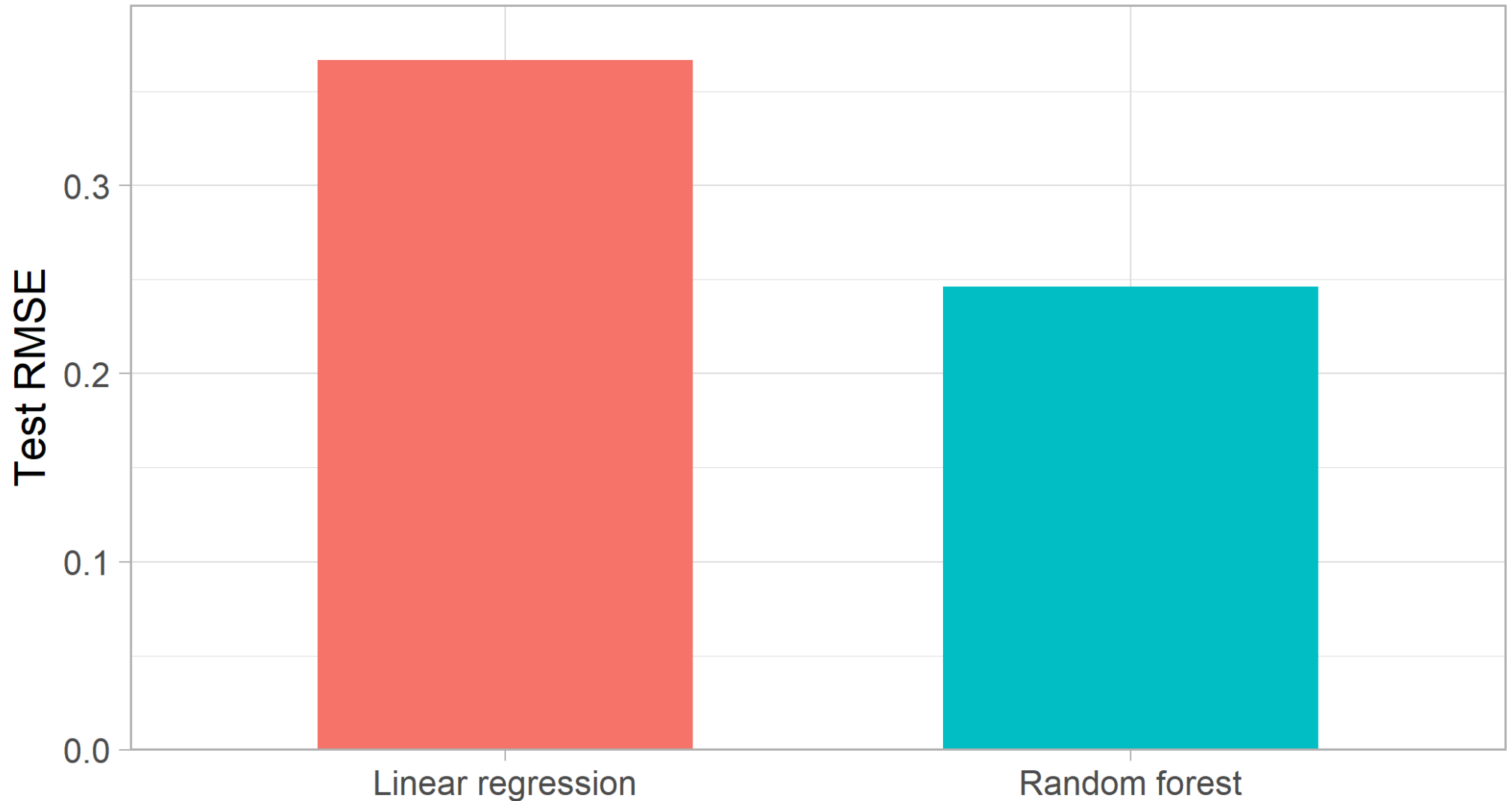
Equal versus Optimized Budget Allocation Across Platforms



Optimized Budget Allocation Across Platform-by-Campaign-Type Combinations



Predictive Accuracy on the Test Set



Conclusion

Overview of Findings

- Platform differences are structural rather than incidental

Application: What This Means for Marketers

- Treat **platform selection** as a strategic decision, not a default choice.
- Match **campaign objective** to the platform rather than relying on platform-wide averages.
- Allocate budget using **marginal return**, not equal splits.
- Use predictive models to support planning when campaign outcomes are nonlinear.

Future Research

- Apply the same framework to **real-world campaign data** rather than simulated data.
- Add **time dynamics** to evaluate how performance shifts across campaigns and seasons.
- Extend the optimization step to include **uncertainty, constraints, and multiple objectives** such as revenue, conversions, and acquisition cost.
- Compare results across additional platforms and broader industry settings.

References

- Ahmad et al. (2018)
- Bekoglu and Onaylı (2016)
- Kaggle Global Ads Performance Dataset

